WAGE DYNAMICS ALONG THE LIFE-CYCLE OF MANUFACTURING PLANTS*

by

Emin Dinlersoz†
Center for Economic Studies
US Census Bureau

Henry Hyatt
Center for Economic Studies
US Census Bureau

and

Sang Nguyen
Center for Economic Studies
US Census Bureau

CES 11-24R March 2013

This paper is a revised version of "Wage Dynamics along the Life-Cycle of Manufacturing Plants" (CES-WP-11-24) from November 2011. A copy of the original paper is available upon request.

The research program of the Center for Economic Studies (CES) produces a wide range of economic analyses to improve the statistical programs of the U.S. Census Bureau. Many of these analyses take the form of CES research papers. The papers have not undergone the review accorded Census Bureau publications and no endorsement should be inferred. Any opinions and conclusions expressed herein are those of the author(s) and do not necessarily represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. Republication in whole or part must be cleared with the authors.

To obtain information about the series, see www.census.gov/ces or contact Fariha Kamal, Editor, Discussion Papers, U.S. Census Bureau, Center for Economic Studies 2K132B, 4600 Silver Hill Road, Washington, DC 20233, CES.Papers.List@census.gov.

ABSTRACT

This paper explores the evolution of average wage paid to employees along the life-cycle of a manufacturing plant in U.S. Average wage starts out low for a new plant and increases along with labor productivity, as the plant survives and ages. As a plant experiences productivity decline and approaches exit, average wage falls, but more slowly than it rises in the case of surviving new plants. Moreover, average wage declines slower than productivity does in failing plants, while it rises relatively faster as productivity increases in surviving new plants. These empirical regularities are studied in a dynamic model of labor quality and quantity choice by plants, where labor quality is reflected in wages. The model's parameters are estimated to assess the costs a plant incurs as it alters its labor quality and quantity in response to changes in its productivity over its life-cycle.

Key Words: Wage dynamics; plant productivity; firm dynamics; plant life-cycle; employment dynamics; manufacturing.

JEL Codes: J31, J21, J24, L60, L23, L11, D24

[*] Any opinions and conclusions expressed herein are those of the author(s) and do not necessarily represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed.

[†] Corresponding author. Center for Economic Studies, U.S. Census Bureau, 4600 Silver Hill Road, Suitland, MD 20746. Tel: (301) 763 7889, Fax: (301) 763 5935, E-mail: emin.m.dinlersoz@census.gov

1 Introduction

How do wages a plant pays to its workers change over time? Are there substantial differences in the time-paths of wages between growing versus failing plants? How are the changes in wages related to productivity movements along the life-cycle of a plant? The answers to these questions matter for theories of wage dynamics emphasizing long-term contracts between workers and firms, human capital evolution within a firm, adjustment costs in factors of production such as labor quality, productivity-wage relationship, and the evolution of organizational complexity with implications on firm workforce hierarchy and composition. This paper provides comprehensive evidence on the evolution of average wage (wage bill per employee) along the life-cycle of a manufacturing plant using data from U.S. Census Bureau's Census of Manufactures over the period 1963-1997. Some facts are presented on how the average wage paid by a plant changes as the plant enters an industry, grows, and ages, and as it approaches exit. The patterns exhibited by wages are tied to the evolution of labor productivity.[1] A dynamic model of plant-level labor quality and quantity adjustment is then built and estimated to explore the asymmetric patterns observed in the evolution of wages for growing versus declining plants.

Empirical evidence on plant age-wage profile is scant. Brown and Medoff (2003) find that older firms tend to pay higher wages. Older firms also offer steeper wage-tenure profiles. Their analysis is based on a relatively small number of workers, and highly established firms with an average age of forty years.[2] Kölling, Schnabel and Wagner (2002) largely confirm Brown and Medoff's (2003) findings using a larger dataset that links establishments to workers in Germany, while Heyman (2007) finds some evidence in favor of a positive firm wage-age relationship using Swedish data. These studies do not track firms or plants over time, and do not consider the joint dynamics of wage and productivity along the firm life-cycle. In addition, while age measures a firm's distance from entry, it does not contain information on its distance to exit, and thus cannot fully account for where a firm stands in its life-cycle. The samples used in these previous studies also do not adequately represent firm age distribution, especially the left tail, which consists of younger plants.

[1] Throughout the paper, the term productivity is used to refer to profitability as measured by revenue per employee.

[2] Their sample consists of 1,067 workers, corresponding to at most as many firms from survey data. See Brown and Medoff (2003) for details.

This paper studies the entire set of manufacturing plants available in the U.S. Census Bureau's Census of Manufactures (CM) over a span of three decades. It characterizes the evolution of wages and labor productivity as plants move away from their entry point, and as they approach exit. New plants start out with lower average wage and productivity compared to established plants, but achieve higher productivity and pay higher average wage as they age. Failing plants, on the other hand, experience declining average wage and productivity. For these plants, average wage does not fall as fast as it rises in the case of surviving new plants. Furthermore, average wage neither rises nor falls as fast as labor productivity does. As a result, failing plants are burdened by an increasingly larger wage bill as a fraction of their revenues, whereas new entrants and young firms incur a smaller wage bill relative to their revenues compared to mature plants. The asymmetric evolutions of labor productivity and average wage along the life-cycle of a plant are the focus.

Given the link between a plant's productivity and its dynamics, it is important to understand how changes in wages are tied to changes in productivity. The model in this paper makes this connection explicit.[3] In the model, a plant's life-cycle dynamics is driven by two stochastic processes; one that drives productivity, and the other the wage rate per unit of labor quality the plant faces. A plant chooses the quality and quantity of its labor force in the presence of these two random elements. The motivation of labor quality, in addition to quantity, is based on prior empirical evidence which suggests that worker quality accounts for a considerable portion of wage differential across plants of different ages.[4] A further motivation is the evidence on a strong positive relationship between a firm's labor quality and productivity.[5] Worker quality can have alternative interpretations, such as human capital, a worker's skill, effort and hours, or the degree of essentiality of a worker in the production process. A higher quality labor force comes at higher wages. Average wage reflects, in part, the quality associated with labor

[3]Some recent studies tie firm-level productivity movements to wage changes using reduced-form econometric models. See, e.g., Fuss and Wintr (2009).

[4]See, e.g., Brown and Medoff (2003) and Kölling, Schnabel and Wagner (2002), Griliches and Regev (1995), and Bahk and Gort (1993). The analysis here does not attempt to disentangle the effect of worker versus plant characteristics on wage dynamics. The goal is not to explain away the differences in wages across plants at different stages of their life-cycles, but rather to explore the link between plant-level productivity dynamics and the evolution of wages along a plant's life-cycle.

[5]See, e.g., Bahk and Gort (1993) and Griliches and Regev (1995).

input. This relationship is also supported by the earlier work of Gort, Sapra, and Bahk (1990) and Gort, Bahk, and Wall (1993), which suggest that variations in average wage can be mainly attributable to differences in human capital rather than differences in the prices of a given labor type. Wages in the model thus have an exogenous component that the plant takes as given, and an endogenous component that depends on average labor quality, which is chosen by the plant. As a plant's productivity increases, the plant desires to increase both its employment and average quality of its labor force, and a decline in productivity induces a downward pressure on both labor quality and employment. Adjustments in labor quality, however, have costs associated with them. These costs may curtail the extent to which average wage can move up or down.

Adjustment costs in labor quantity and quality broadly represent, in a reduced form, the effects of a variety of considerations, including frictions in reorganizing production (e.g. plant expansion and contraction) and frictions in the labor market (e.g. hiring and firing), the effects of institutions (e.g. unions), and regulations.[6] Exogenous shocks to wages can induce a plant to alter its wages at a cost. Some workers can exercise their outside options if their wages are not adjusted when demand for their services rises in the economy. Plants experiencing fast productivity growth may need to reorganize their workforce and increase the average worker quality, and hence, output. Such reorganization may involve creating new jobs, training, and moving labor across ranks and tasks. Similarly, as a plant experiences persistent episodes of low productivity, it may shed some of its employees non-essential for the main activity of the plant, and it may even reduce compensation in real terms for some workers.[7] Some workers may also quit in anticipation of exit, and the plant may then have to reorganize its labor force to maintain production. All of such events require costly adjustments to average wage. Unions, minimum wage laws, and other regulations can also limit reductions in real wages.

To assess adjustment costs associated with average wage, the model's parameters are estimated. The estimates reveal that labor quality, inferred from wages and revenue, is an important input of production. Furthermore, there are statistically and economically significant

[6] Unions in manufacturing, while weaker recently, were effective for much of the sample period used in this paper that goes back to 1963.

[7] There need not be a reduction in nominal compensation. If a plant freezes nominal wages, real wages would fall when there is inflation.

asymmetric adjustment costs. The average annual cost of average wage adjustment in any direction constitutes up to 1.6% of a plant's revenue at the median of the adjustment cost distribution, when only the continuing plants are considered. For plants adjusting average wage downwards, the annual adjustment cost claims, on average, up to 3.6% of revenue. For plants adjusting average wage upwards, the annual adjustment cost makes up as much as 0.8% of revenue on average. These shares are higher when only the exiting plants are considered: 3%, 6.6%, 1.2%, respectively. A version of the model with both average wage and employment adjustment costs is also estimated. A plant may not be able to adjust its wages holding employment constant. Similarly, changes in employment can be used to induce changes in wages, through a change in the composition of a plant's labor force. Using both channels of adjustment can help address biases due to using only one.

This paper contributes in a number of dimensions to the analysis of wage dynamics. First, it reveals a new set of facts pertaining to wage dynamics along a plant's life-cycle. Second, it explores the role of adjustment costs associated with a plant's average wage. Third, it considers the dynamic interaction between labor quality and quantity at the plant level. This interaction has implications for the time-paths of wages and employment, and for adjustment costs associated with both. Finally, the data used here is much richer than those in most previous studies.[8] Furthermore, prior studies do not analyze how adjustment costs change along the plant or firm life-cycle.

The rest of the paper is organized as follows. Section 2 describes the data and presents some observations regarding the evolution of wages along a plant's life-cycle. Section 3 introduces the model. The estimation methodology is presented in Section 4. The estimation results are discussed in Section 5. Section 6 offers concluding remarks.

[8] Most prior work focus on employment and capital adjustment. Studies at the firm level either use relatively small samples or non-representative samples from U.S. firm population, such as relatively large publicly-traded firms in S&P's Compustat Database, whereas other studies use aggregate data, which may hide patterns at the micro level. Examples include, among others, Alonso-Borrego (1998), Shapiro (1986), Pfann and Palm (1993), Hall (2004), and Merz and Yashiv (2007).

2 The evolution of wages

2.1 Data

The primary data source is the Longitudinal Research Database (LRD), maintained at the Center for Economic Studies of the US Census Bureau. The LRD describes aspects of manufacturing plants' production, including key plant-level variables such as total value of shipments and value added, as well as employment and total wage bill. The LRD also contains information on classification and identification of plants, such as plants' ownership, location, and industry, as well as various status codes that identify birth, death, and ownership changes. These identifying codes are used in developing the longitudinal plant linkages.[9]

The analysis focuses on a subset of the LRD that includes eight waves of the Census of Manufactures (CM): 1963, 1967, 1972, 1977, 1982, 1987, 1992 and 1997. The focus on these years are driven by the fact that plants are classified consistently into SIC industry codes, for which there was a substantial revision and transition into NAICS industry codes in 2002. The number of plants in the CM range from 305,691 in 1967 to 400,036 in 1997. Using permanent plant numbers, plants were linked from these CM's to form an unbalanced panel for the period 1963-1997. Plant entry, exit, and continuation were identified. A plant is observed at most eight times, when it appears in all waves of the CM.

The key variables for the analysis are constructed as follows. A plant's revenue is its value of shipments deflated to 1987 dollars using 4-digit SIC level industry price deflators from NBER-CES Manufacturing Industry Database based on 1987 SIC code definitions.[10] Deflated value added is also calculated as an alternative measure of revenue, which is used later in the model's estimation. Employment is a plant's total number of workers engaged in production and non-production activities. The main wage variable, the total wage bill, is deflated to 1987 dollars using CPI from Bureau of Labor Statistics. Using a plant's deflated revenue, its deflated wage bill, and its employment, three ratios were constructed for each plant-CM wave observation: (a) average wage – the ratio of wage bill to employment, (b) labor productivity – the ratio of revenue to employment, and (c) the ratio of wage bill to revenue, equivalent to the ratio of (a) to (b). The goal of the empirical analysis to follow is to describe the life-cycle evolutions of

[9] For further information on the LRD, visit http://www.census.gov/ces/dataproducts/economicdata.html.

[10] NBER/CES database is available at http://www.nber.org/data/nbprod2005.html.

these three ratios.

2.2 Main findings

The life-cycle effects on key plant-level variables are estimated using several specifications of the following OLS regression

$$Y_{it} = \alpha + \sum_{\tau} \beta_X^{\tau} X_{it}^{\tau} + \sum_{\tau} \beta_E^{\tau} E_{it}^{\tau} + \boldsymbol{\beta}_Z' \mathbf{Z}_{it} + f_t^I + \varepsilon_{it}, \tag{1}$$

where i indexes plants, t indexes census years, Y_{it} is either average wage, labor productivity, or wage-bill-to-revenue ratio, X_{it}^{τ} is an indicator of whether a plant is $\tau \in \{0, 5, 10, 15, 20\}$ years *to* its exit point (the last census it is observed), E_{it}^{τ} is an indicator of whether a plant is $\tau \in \{0, 5, 10, 15, 20\}$ years away *from* its entry point (the first census it is observed), and $\mathbf{Z}_{it}$ is a vector of plant-level controls.[11,12] Industry-year fixed effects, f_t^I, were also added to control for time and industry specific effects, such as the effects of industry life-cycles and aggregate shocks. The omitted category, referred to as mature plants, contains the plants that are more than twenty years away from their entry or exit, and the plants whose entry or exit dates do not fall into the sample period.[13] The error term ε_{it} is assumed to have an unobserved plant-specific component, and errors are clustered by plant.

Of particular interest are the life-cycle indicators, X_{it}^{τ} and E_{it}^{τ}. They track the evolution of the dependent variable Y_{it} as a plant moves away from entry into maturity, and as it approaches exit. The coefficients β_X^{τ} and β_E^{τ} quantify the magnitudes of these life-cycle effects. β_X^{τ} measures how much being τ years away from exit matters for Y_{it}, controlling for time from entry as identified by the indicators E_{it}^{τ}, in addition to other controls. Similarly, β_E^{τ} measures how much being τ years from entry matters for Y_{it}, controlling for time to exit identified by X_{it}^{τ}, in addition to other controls.

Specification I in Table I generates Figure I. Consider, first, the evolution of average wage in Figure Ia. New plants begin with an average wage roughly \$2,700 lower than that of mature plants. It takes at least twenty years for this gap to drop below \$300. On the other hand,

[11] The total employment of a plant was restricted to the range 5 to 10,000 employees. In addition, the top and bottom percentiles of the three dependent variables were trimmed to reduce the influence of any outliers.

[12] Note that some plants indicated by a given E_{it}^{τ} may also be in the set of plants indicated by some $X_{it}^{\tau\prime}$s.

[13] The regression specifications are similar to those used in Foster, Haltiwanger, and Syverson (2008).

average wage starts to fall as a plant approaches exit. Plants twenty years away from exit have about $200 lower average wage than mature plants. This gap grows to $2,000 by the time of exit. Wages tend to rise faster for surviving entrants than they fall for plants approaching exit, as indicated by the slopes of the two average wage profiles for the parts of the life-cycle on either side of the y-axis.

Next, turn to the evolution of labor productivity in Figure Ib. Compared to mature plants, new plants start off with approximately $5,700 productivity disadvantage that diminishes over time. Failing plants exhibit a much larger productivity disadvantage. Exiting plants have about $10,000 lower productivity, and even as early as ten years prior to exit, plants that eventually exit have around $4,000 lower productivity. Labor productivity tends to fall faster for plants nearing exit than it rises for aging entrants.

The findings in Figures Ia-b together suggest the evolution of the ratio of a plant's wage bill to its revenue, equivalent to the fraction of a worker's productivity provided to the worker in the form of wages. The estimated life-cycle pattern for this ratio is shown in Figure Ic. New plants spend a lower fraction of their revenue on wage bill compared to mature plants, although this wage advantage largely disappears over the next fifteen years for surviving plants. Plants approaching exit, however, exhibit a small but statistically significant increase in this ratio. Failing plants on average spend a higher fraction of their revenue on wages compared to the omitted category.

The findings largely continue to hold when other controls, $\mathbf{Z}_{it}$, are added. Specification II includes a cubic polynomial in plant size, which has been found to be highly positively associated with wages (see, e.g., Brown and Medoff (1989)), in addition to an indicator of whether the plant is part of a multi-unit firm, and location (Metropolitan Statistical Area) fixed effects. Plant size is measured by total employment. While the magnitudes of life-cycle effects in average wage are now somewhat smaller in absolute value, their signs and significance resemble those in specification I. The most important difference is in the case of labor productivity for young plants. Compared to specification I, young plants now exhibit a much faster productivity growth, and seem to wipe out their productivity disadvantage vis á vis mature plants by their fifth year after entry. As a result, wage-bill-to-revenue ratio for entering plants is even lower, and stays lower longer as they age.

The overall theme of Table I can be summarized as follows. Average wage is low for entering

plants, and gradually approaches that of mature plants as plants age. As plants get closer to exit, wages start to fall on average, but not as fast as they increase for surviving and aging entrants. Labor productivity is also much lower for entering plants, but improves as they age. Plants nearing exit have a more substantial productivity disadvantage, visible even twenty years prior to exit. Griliches and Regev (1995) find a similar pattern in an early study of Israeli firms and dub this effect "the shadow of death": firms that are due to exit in the future are less productive in the present. Wages rise slower than labor productivity does for young plants as they survive and age, and they fall slower than labor productivity does for failing plants. However, the relative pace of wage and productivity growth in young plants differs from the relative pace of decline in failing plants. This asymmetry manifests itself in the evolution of wage-bill-to-revenue ratio. For young plants, wages constitute a smaller fraction of revenue compared to mature plants, whereas plants approaching exit are burdened with a wage bill that claims a larger fraction of their revenue compared to mature plants.

Evolutions depicted in Figure I can be interpreted as the typical or average evolutions that would take place if a plant goes through all stages of the life-cycle. One concern is a potential composition bias in these evolutions due to differences in plant life-span. For instance, short-lived plants may be a special group, perhaps those that are born with a much lower productivity and live for a period of at most 2 or 3 censuses, and experience a monotonic decline in average wages and productivity over that period until exit. As a result, the latter part of the life-cycle trends in Figure I may be driven solely by such plants, instead of being shared by longer-lived plants. As a further robustness check for the patterns in Figure I, and as an alternative to the panel analysis pooling all plants, the evolution of average wage is plotted in Figure II along the life-span of plants that are born within the sample period and live for a given number of censuses, a minimum of 2, and a maximum of 6. This longitudinal analysis ensures that the same set of plants are observed over time for any given life-span. The estimated life-cycle effects on average wage plotted in Figure II are obtained using a similar regression specification to (1) with only a set of dummies that indicate the number of years from entry interacted by a life-span dummy. The full set of results for this exercise are in Table AI in Appendix.[14] The inverted-U shaped pattern in Figure Ia also emerges for all different life-spans in Figure II. Even

[14]Note that time-to-exit indicators are excluded in these regressions, as including them together with time-from-entry indicators would lead to collinearity with life-span indicator.

relatively short-lived plants appear to experience an increase in average wage after entry, and a decline before exit. Note also from Table AI that wages decline slower than productivity does as plants approach to exit, similar to the pattern found in Table I. The results thus appear to be robust to life-span considerations.

3 The model

The model is motivated by the empirical findings, which suggest an asymmetric evolution of average wage for plants experiencing productivity increase versus decline. It explores the potential role of adjustment costs in dampening the changes in average wage across the life-cycle of a plant, especially when a plant experiences productivity decline and approaches exit. Consider an industry with many plants. Both the quantity, L, and quality, q, of labor are factors of production for a plant. Worker quality, q, can have alternative interpretations, such as a worker's skill level, human capital, effort (including hours worked), or the degree of essentiality of a worker in production. Plants can be price-takers in their output market, or local monopolies which set prices given their downward sloping demand. There is an infinite number of discrete time periods. Plants receive random shocks to their productivity each period. There are also shocks to an exogenous wage rate, w, per unit of labor quality, which a plant takes as given. This wage rate is allowed to be plant-specific, potentially reflecting the plant's local labor market as well as the plant's characteristics. There is a large number of workers with varying quality levels available for hire at the wage rate w. For a plant with employment L, workers are indexed on the interval $[0, L]$, and the quality of each worker $l \in [0, L]$ is denoted by $q(l)$. There are two distinct channels through which a plant's average wage can change. First, a plant has to adjust its wages in response to changes in the wage rate per unit of worker quality. The second channel is the ability of a plant to alter its wage bill through adjustment of its average worker quality.

3.1 Adjustment cost for wages

A plant chooses the quantity and average quality of its labor force, as well as other inputs, to generate output by using a Cobb-Douglas production technology. The period profit of a

plant is

$$\Pi(q_t, L_t; q_{t-1}, L_{t-1}, w_t, w_{t-1}, \theta_t) = \theta_t L_t^{\alpha} q_t^{\gamma} - w_t q_t L_t - A(q_t, L_t; q_{t-1}, L_{t-1}, w_t, w_{t-1}) - F. \quad (2)$$

In the production function represented by the first term on the right hand side of (2), L_t is the quantity of labor, q_t is the average quality per unit of labor defined by $q_t = \frac{1}{L_t}\int_0^{L_t} q(l)dl$, and $\theta_t \geq 0$ is a productivity shock that includes aggregate and idiosyncratic shocks, as well as the output price. The production function exhibits decreasing returns, $\alpha + \gamma < 1$.[15] This formulation allows different elasticities of output with respect to labor quality and quantity.[16]

Costs are represented by the remaining terms on the right hand side of (2). The first component of the cost is the total wage bill, $w_t q_t L_t$, where w_t is the wage per unit of labor quality. The second component is the adjustment cost

$$A(q_t, L_t; q_{t-1}, L_{t-1}, w_t, w_{t-1}) = \left[I_t^U \frac{\lambda^U}{2} + I_t^D \frac{\lambda^D}{2} \right] \left(\frac{w_t q_t - w_{t-1} q_{t-1}}{w_{t-1} q_{t-1}} \right)^2 w_{t-1} q_{t-1} L_{t-1}, \quad (3)$$

where $\lambda^U \geq 0$ and $\lambda^D \geq 0$ are the parameters of the quadratic adjustment costs associated with upward and downward adjustments in average wage, $w_t q_t$, indicated by $I_t^U \equiv I(w_t q_t > w_{t-1} q_{t-1})$ and $I_t^D \equiv I(w_t q_t \leq w_{t-1} q_{t-1})$, respectively. The base to which adjustment in average wage applies is the total wage bill in period $t-1$, $w_{t-1} q_{t-1} L_{t-1}$. Changes in average wage has two sources: changes in w_t, and the plant's choice of q_t. When w_t changes, the plant has to incur an adjustment cost in wages even when it does not alter the average quality of its labor force.[17] Finally, F is a fixed cost of operation that is avoidable only if the plant exits.

The distribution of θ_t is given by the *c.d.f.* $H_\tau(\theta_t|\theta_{t-1}) \equiv H(\theta_t|\theta_{t-1}, \tau)$, which specifies the general dependence of a period's productivity shock on its previous value, and plant age,

[15] Suppose the plant is a price-taker and the underlying production function is given by $\widetilde{\theta}_t \left(M_t^{1-\widetilde{\alpha}-\widetilde{\gamma}} L_t^{\widetilde{\alpha}} q_t^{\widetilde{\gamma}} \right)^{\delta}$, where M_t is some other input that can be costlessly adjusted, $\widetilde{\alpha}, \widetilde{\gamma} \in (0,1)$ are the shares of labor quantity and quality in output such that $\widetilde{\alpha} + \widetilde{\gamma} < 1$, and $\delta \in (0,1)$ reflects decreasing returns at the plant level to some fixed input (such as capital or physical space). If the unit price of M_t is m_t, the optimization only with respect to M_t leads to the production function in (2) where $\alpha = \frac{\delta\widetilde{\alpha}}{1-\delta(1-\widetilde{\alpha}-\widetilde{\gamma})}$, $\gamma = \frac{\delta\widetilde{\gamma}}{1-\delta(1-\widetilde{\alpha}-\widetilde{\gamma})}$, and θ_t is a function of $\widetilde{\theta}_t, \widetilde{\alpha}, \widetilde{\gamma}, \delta, m_t$ and output price p_t. A similar interpretation applies if a plant has local market power, in which case δ is a function of the elasticity of demand.

[16] When there are no adjustment costs, the elasticity of output with respect to labor quantity and quality must be the same ($\alpha = \gamma$) for interior maximizers to exist in the firm's optimization problem. The presence of adjustment costs and dynamic effects of adjustments breaks this constraint, and allows interior maximizers in cases when $\alpha \neq \gamma$.

[17] The two different sources of adjustment are not separately identified in the data available for this study.

τ.[18] The life-cycle effects on productivity are assumed to be driven by the process $H_\tau(\theta_t|\theta_{t-1})$. There are several processes that can generate a life-cycle evolution of productivity similar to the dynamics pictured in Figure Ib. One specific example is as follows. Suppose that new entrants start with an initial productivity drawn from the distribution $H_1(\theta)$. After a plant's first period in the industry, $H_\tau(\theta_t|\theta_{t-1})$ is strictly decreasing in θ_{t-1} for all $\tau > 1$. In addition, for some θ^u, $H_\tau(\theta_t|\theta_{t-1})$ is strictly decreasing in $\tau \geq 1$ as long as $\theta_{t-1} \geq \theta^u$, and for some $\theta^l < \theta^u$, it is strictly increasing in $\tau \geq 1$ as long as $\theta_{t-1} \leq \theta^l$. Entrants thus start with a lower productivity on average than successful incumbents. Entrants that have better shocks are more likely to experience even better shocks in the future. If an entrant's productivity ever exceeds the threshold θ^u, the productivity increases on average for all periods thereafter as long as productivity stays above θ^u.[19] However, if a plant ever experiences a productivity level below θ^l, future productivity draws become increasingly more unfavorable on average. Such a process can generate the gradual decline in average productivity prior to exit, as well as the gradual increase in average productivity after entry. The exact form of the productivity process is not specified, as such a specification is not needed for the model's estimation. The distribution of the wage rate, w_t, is also given by a $c.d.f.$ $G(w_t|w_{t-1})$.

Denote the state variable for a plant by $\mathbf{s}_t = (q_{t-1}, L_{t-1}, w_t, w_{t-1}, \theta_t)$. The plant makes its decisions after observing θ_t and w_t. The exit decision is denoted by the discrete choice X_t such that $X_t = 1$ if the plant exits, and $X_t = 0$ if the plant continues. The value of a plant is

$$V(\mathbf{s}_t) \equiv \max_{X_t, L_t, q_t} (1 - X_t) \left(\Pi(q_t, L_t; \mathbf{s}_t) + \beta E\left[V(\mathbf{s}_{t+1})\right]\right), \tag{4}$$

where β is the discount factor and the plant's exit value is normalized to zero. Given the assumptions of the model so far, dynamic programming arguments in Stokey and Lucas (1989) guarantee the existence of a value function V in (4), which is strictly increasing in θ_t. Because V is monotonic in θ_t, exit occurs the first time θ_t is such that $V(\mathbf{s}_t) \leq 0$.

The entry process is as follows. At the beginning of every period, there is a large number (a continuum) of ex-ante identical potential entrants which can enter the industry by paying a

[18] Several studies, including some recent ones such as Abraham and White (2006), find high persistence in productivity at the plant level.

[19] Suppose that $\lim_{\tau\to\infty} H_\tau(\theta_t|\theta'_{t-1}) = \overline{H}(\theta_t|\theta_{t-1})$ such that for all $\theta_t > 0$, $\overline{H}(\theta_t|\theta_{t-1}) > 0$, ensuring that even very small shocks have a positive probability of occurring in the limit, so that no firm escapes to a no exit region.

sunk entry cost of $\kappa > 0$. Upon entry, an entrant's initial productivity draw comes from the distribution $H_1(\theta)$. The expected value from entry in any period is $V_t^e = E[V(\mathbf{s}_t)]$, where an entrant's initial state $\mathbf{s}_t = (0, 0, w_t, w_{t-1}, \theta_t)$ reflects the fact that its prior labor quality and quantity are both zero.

Let V_i denote the derivative of V with respect to its ith argument when it exists.[20] At an interior solution, a continuing plant's choice of L_t satisfies

$$\alpha\theta_t L_t^{\alpha-1} q_t^{\gamma} - w_t q_t + \beta E\left[V_2(\mathbf{s}_{t+1})\right] = 0, \tag{5}$$

where

$$E\left[V_2(\mathbf{s}_{t+1})\right] = -E\left[\left(I_{t+1}^U \frac{\lambda^U}{2} + I_{t+1}^D \frac{\lambda^D}{2}\right)\left(\frac{w_{t+1}q_{t+1} - w_t q_t}{w_t q_t}\right)^2\right] w_t q_t. \tag{6}$$

Similarly, a continuing plant's choice of q_t satisfies

$$\gamma\theta_t L_t^{\alpha} q_t^{\gamma-1} - w_t L_t - \left[I_t^U \lambda^U + I_t^D \lambda^D\right]\left(\frac{w_t q_t - w_{t-1}q_{t-1}}{w_{t-1}q_{t-1}}\right) w_t L_{t-1} + \beta E\left[V_1(\mathbf{s}_{t+1})\right] = 0, \tag{7}$$

where

$$E\left[V_1(\mathbf{s}_{t+1})\right] = E\left[\left(I_{t+1}^U \frac{\lambda^U}{2} + I_{t+1}^D \frac{\lambda^D}{2}\right)\frac{L_t}{w_t^2 q_t^2}\left(w_{t+1}^2 q_{t+1}^2 - w_t^2 q_t^2\right) w_t\right]. \tag{8}$$

The first order conditions (5) and (7) implicitly determine a plant's labor quantity and quality policies $L(\mathbf{s}_t)$ and $q(\mathbf{s}_t)$. Let $\Pi^*(\mathbf{s}_t)$ be the period profit evaluated at optimal policies $L(\mathbf{s}_t)$ and $q(\mathbf{s}_t)$. The exit policy is then

$$X(\mathbf{s}_t) = \begin{cases} 0 & \text{if } \Pi^*(\mathbf{s}_t) + \beta E\left[V(\mathbf{s}_{t+1})\right] > 0, \\ 1 & \text{otherwise.} \end{cases}$$

Using (5), the wage bill can be expressed as a percentage of revenue as

$$f_t = \frac{L_t w_t q_t}{\theta_t L_t^{\alpha} q_t^{\gamma}} = \alpha + \beta \frac{E\left[V_2(\mathbf{s}_{t+1})\right] L_t}{\theta_t L_t^{\alpha} q_t^{\gamma}}. \tag{9}$$

Using (6) in (9) and rearranging, one obtains

$$f_t = \frac{\alpha}{1 + \beta a_{t+1}}, \tag{10}$$

where

$$a_{t+1} = E\left[\left(I_{t+1}^U \frac{\lambda^U}{2} + I_{t+1}^D \frac{\lambda^D}{2}\right)\left(\frac{w_{t+1}q_{t+1} - w_t q_t}{w_t q_t}\right)^2 \middle| \theta_t, w_t\right], \tag{11}$$

[20] The derivative of V with respect to q_t does not exist at $q_t = \frac{w_{t-1}}{w_t} q_{t-1}$ when $\lambda^U \neq \lambda^D$.

is the expected value of the adjustment cost as a percentage of the period t wage bill. If the adjustment costs are zero ($\lambda^U = \lambda^D = 0$), the labor's share of revenue is $f_t = \alpha$, the static elasticity of output with respect to labor quantity. An implication of (10) is that higher expected adjustment in average wage in the next period is accompanied by a lower share of labor in current period revenue ($f_t < \alpha$).[21] Consider now the case where the policies $L(\mathbf{s}_t)$ and $q(\mathbf{s}_t)$ are both strictly increasing in θ_t.[22] A high (low) productivity shock then implies an upward (downward) adjustment in both L and q. When $\lambda^U < \lambda^D$, a plant that experiences an unfavorable productivity shock adjusts q downward, but not as much as a plant that gets a favorable shock and adjusts q upward. Average wage in declining plants would then decrease more slowly than it rises in growing plants. Thus, a_{t+1} can be lower for a plant adjusting downward, implying a higher f_t for plants experiencing decline in productivity. If a growing plant makes increasingly smaller expected adjustments to average wage, then a_{t+1} declines over time, and f_t increases, as in Figure Ic. On the other hand, if adjustments for plants which experience persistent productivity decline become increasingly small as they approach exit, then a_{t+1} also declines as a plant nears exit, and f_t continues to rise as in Figure Ic.

3.2 Adjustment costs for wages and employment

Consider now an adjustment cost for employment, in addition to that in average wage. Incorporating both adjustments costs can account for some of the potential bias in the measurement of these two costs. Labor is not homogeneous. A change in employment can rarely be accomplished holding average worker quality constant, and vice versa. The adjustment cost in (3) is now modified as

$$\begin{aligned} A(q_t, L_t; q_{t-1}, L_{t-1}, w_t, w_{t-1}) &= \left[I_t^U \tfrac{\lambda^U}{2} + I_t^D \tfrac{\lambda^D}{2}\right] \left(\tfrac{w_t q_t - w_{t-1} q_{t-1}}{w_{t-1} q_{t-1}}\right)^2 w_{t-1} q_{t-1} L_{t-1} \\ &\quad + \left[J_t^U \tfrac{\nu^U}{2} + J_t^D \tfrac{\nu^D}{2}\right] \left(\tfrac{L_t - L_{t-1}}{L_{t-1}}\right)^2 L_{t-1}, \end{aligned} \tag{12}$$

where $\nu^U \geq 0$ and $\nu^D \geq 0$ are the parameters of the quadratic adjustment cost associated with upward and downward employment adjustments, indicated respectively by $J_t^U \equiv I(L_t > L_{t-1})$

[21] This result follows because the plant's current wage bill and its expected cost of adjustment in wages next period must be compensated by the labor's share of current revenue.

[22] Such monotonicity holds under certain restrictions on the parameters of the model.

and $J_t^D \equiv I(L_t \leq L_{t-1})$. The specification of adjustment costs takes into account two margins of adjustment: an intensive margin that applies only to average wage, and an extensive margin that applies only to employment. These two margins are interrelated. Adjustment in average wage has an effect on a plant's current and future employment, and adjustment in employment influences current and future average wage.

For a continuing plant, the first order condition for q_t is the same as (7). For L_t, the first order condition now becomes

$$\alpha\theta_t L_t^{\alpha-1} q_t^{\gamma} - w_t q_t - \left[J_t^U \nu^U + J_t^D \nu^D\right] \left(\frac{L_t - L_{t-1}}{L_{t-1}}\right) + \beta E\left[V_2(\mathbf{s}_{t+1})\right] = 0. \tag{13}$$

3.3 Alternative growth rates

As an alternative, adjustment costs which allow for symmetric and bounded growth rates can also be considered. These alternative specifications for average wage and employment are given by replacing the denominators of the squared terms in (3) and (12) with $\frac{1}{2}(w_t q_t + w_{t-1} q_{t-1})$ and $\frac{1}{2}(L_t + L_{t-1})$, respectively. These specifications restrict the growth rates in average wage or employment to the interval $[-2, 2]$, and are robust to outliers and to potential biases due to mean reversion in plant size and wage bill.

4 Estimation

4.1 Adjustment costs for average wage

Consider first the model with adjustment cost for average wage only. Condition (7) can be rewritten after multiplying through by q_t as

$$\gamma\theta_t L_t^{\alpha} q_t^{\gamma} - w_t q_t L_t - \left[I_t^U \lambda^U + I_t^D \lambda^D\right] \left(\frac{w_t q_t - w_{t-1} q_{t-1}}{w_{t-1} q_{t-1}}\right) w_t q_t L_{t-1} + \beta E\left[V_1(\mathbf{s}_{t+1})\right] q_t = 0. \tag{14}$$

Multiplication by q_t ensures that average wage, $w_t q_t$, appears in the first order condition, rather than just w_t. The former can be calculated using a plant's wage bill and employment, whereas the latter is not observed. Note that the labor quality q_t is also unobserved. Consequently, the parameters γ, λ^U, and λ^D, are identified from wages and output. The implicit assumption is that part of the wages at the plant level is proportional to labor quality. This approach differs from those that use direct measures of labor quality based on observable worker characteristics,

such as education and experience. The estimation of parameters associated with labor quality here does not require a specific measure or index of labor quality, or the estimation of an earnings function.[23]

To implement an estimation based on the first order conditions, assume that these conditions are not exactly fulfilled, but hold subject to a non-systematic error that stem from several sources, such as optimization errors or the differences between anticipated and realized output price, or wage rate per unit of quality. These idiosyncratic errors are assumed to be randomly distributed over plants. Following Hansen and Singleton (1982), the ex-post error can be expressed, using (8) and (14), as a function of the parameters $\Phi = \{\alpha, \gamma, \lambda^U, \lambda^D\}$

$$\begin{aligned} \varepsilon_t(\Phi) &= -\gamma\theta_t L_t^{\alpha} q_t^{\gamma} + w_t q_t L_t + \left[I_t^U \lambda^U + I_t^D \lambda^D\right] \left(\frac{w_t q_t - w_{t-1} q_{t-1}}{w_{t-1} q_{t-1}}\right) w_t q_t L_{t-1} \\ &\quad -\beta \left[I_{t+1}^U \frac{\lambda^U}{2} + I_{t+1}^D \frac{\lambda^D}{2}\right] \frac{L_t}{w_t q_t} \left(w_{t+1}^2 q_{t+1}^2 - w_t^2 q_t^2\right), \end{aligned} \tag{15}$$

The ex-post error in labor choice, after using (5) and (6), is

$$\eta_t(\Phi) = -\alpha\theta_t L_t^{\alpha-1} q_t^{\gamma} + w_t q_t + \beta \left[I_{t+1}^U \frac{\lambda^U}{2} + I_{t+1}^D \frac{\lambda^D}{2}\right] \left(\frac{w_{t+1} q_{t+1} - w_t q_t}{w_t q_t}\right)^2 w_t q_t. \tag{16}$$

For exiting plants, the decision variables are not observed for the period after their exit. However, the event of exit contains additional information, as exit probability depends on the parameters of interest

$$\Pr(X(\mathbf{s}_t) = 1) = \Pr(V(\mathbf{s}_t) \leq 0) = \Pr(\theta_t < \theta^*(q_{t-1}, L_{t-1}, w_t, w_{t-1})),$$

where $\theta^*(q_{t-1}, L_{t-1}, w_t, w_{t-1}, \theta_{t-1})$ is the threshold productivity shock such that a plant with $\mathbf{s}_t = (\theta_t, q_{t-1}, L_{t-1}, w_t, w_{t-1})$ and $\theta_t < \theta^*$ exits. Pakes (1994) shows that one can substitute the discrete exit policy $X(\mathbf{s}_t)$ into the expected discounted future profits, and proceed as in the case of continuing plants. Thus, for a plant that continues from time $t+1$ to $t+2$, $X(\mathbf{s}_{t+1}) = 0$ and the ex-post errors are as defined earlier. For a plant that exits at time $t+1$, $X(\mathbf{s}_{t+1}) = 1$ and the ex-post errors become

$$\varepsilon_t(\Phi) = -\gamma\theta_t L_t^{\alpha} q_t^{\gamma} + w_t q_t L_t + \left[I_t^U \lambda^U + I_t^D \lambda^D\right] \left(\frac{w_t q_t - w_{t-1} q_{t-1}}{w_{t-1} q_{t-1}}\right) w_t q_t L_{t-1}, \tag{17}$$

$$\eta_t(\Phi) = -\alpha\theta_t L_t^{\alpha-1} q^{\gamma} + w_t q_t. \tag{18}$$

[23]For instance, Griliches and Regev (1995) offer a link between labor productivity and a labor quality index within the framework of a reduced form linear specification of labor productivity.

Equations (15)-(18) can be used in a generalized method of moments (GMM) framework to estimate the parameters $\Phi = \{\alpha, \gamma, \lambda^U, \lambda^D\}$. Consider moment conditions of the form

$$\begin{aligned} m_i^{\varepsilon}(\Phi) &= E[z_{it}\varepsilon_t(\Phi)] = 0, \\ m_j^{\eta}(\Phi) &= E[z_{jt}\eta_t(\Phi)] = 0, \end{aligned} \tag{19}$$

where z_{it}, z_{jt} denote instrumental variables $i = 1, ..., I$, $j = 1, ..., J$. The GMM estimation can be carried out using the empirical counterparts of the $I + J$ moment conditions in (19). The estimation involves endogenous variables (revenue, employment, and average wage) and has four unknown parameters, and two equations that determine ex-post error terms. At least four instruments are needed to estimate the four parameter version of the model. The model suggests that the lagged values of decision variables, revenues, value added, and wage bill are predetermined in period t, and hence are orthogonal to both error terms, $\varepsilon_t(\Phi)$ and $\eta_t(\Phi)$. Any of these variables and their functions can serve as an instrument in both types of moment conditions in (19). The instruments used thus consist of values of revenue, value added, wage bill, average wage, their squares, and their interactions.[24] It is important to include exiting plants in the estimation as not doing so can induce bias in both the production function and adjustment cost parameter estimates. The estimation is done separately for continuing plants and all plants (continuing and exiting) to assess this bias. For each census year t, continuing plants are the ones that are observed consecutively in years $t - 5$, t, and $t + 5$, for $t = 1967, ..., 1992$.[25] Similarly, for each census year t, all plants include continuing plants and the exiting plants – those plants that are observed in census years $t - 5$ and t, but not in $t + 5$. In the estimation, the discount factor is set to $\beta = 0.78$ for the quinquennial data, which corresponds to a 95% discount rate for annual data. The estimation is repeated for the case of adjustment cost with alternative growth rate using the corresponding first order conditions.[26]

[24] The results were not significantly different when twice-lagged versions of the variables were also used as instruments. Note also that there are more moment conditions than parameters, so the model is over-identified.

[25] With the exception that for $t = 1967$ the previous census year is 1963, corresponding to $t-4$.

[26] The first order conditions for the case with alternative growth rates are omitted to save space and are available upon request.

4.2 Adjustment costs for average wage and employment

For the case with both employment and average wage adjustment cost, the ex-post error for L_t is

$$\begin{aligned}\eta_t(\Phi) &= -\alpha\theta_t L_t^{\alpha-1} q_t^{\gamma} + w_t q_t + \left[J_t^U \nu^U + J_t^D \nu^D\right] \left(\tfrac{L_t - L_{t-1}}{L_{t-1}}\right) \\ &+ \beta \left[I_{t+1}^U \tfrac{\lambda^U}{2} + I_{t+1}^D \tfrac{\lambda^D}{2}\right] \left(\tfrac{w_{t+1}q_{t+1} - w_t q_t}{w_t q_t}\right)^2 w_t q_t \\ &- \beta \left[J_{t+1}^U \tfrac{\nu^U}{2} + J_{t+1}^D \tfrac{\nu^D}{2}\right] \left(\tfrac{L_{t+1}^2 - L_t^2}{L_t^2}\right).\end{aligned} \tag{20}$$

The ex-post error for q_t is the same as (15).

For exiting plants, the ex-post error for L_t is

$$\eta_t(\Phi) = -\alpha\theta_t L_t^{\alpha-1} q^{\gamma} + w_t q_t + \left[J_t^U \nu^U + J_t^D \nu^D\right] \left(\frac{L_t - L_{t-1}}{L_{t-1}}\right),$$

and the ex-post error for q_t is the same as (17). The parameters to be estimated are now $\Phi = \{\alpha, \gamma, \lambda^U, \lambda^D, \nu^U, \nu^D\}$. The GMM estimation is carried out for the six parameter version of the model similar to the case with four parameters. In this case, at least six instruments are needed, and the instruments described in the previous section are used.

5 Results

5.1 Parameter estimates

The GMM estimates of parameters using the entire sample of manufacturing plants are reported in Table II. Two measures of revenue are used alternatively: deflated total value of shipments (top panel) and deflated value added (bottom panel). Adjustment costs with both the conventional and the alternative growth rates are considered. The estimates in specifications I and III pertain only to continuing plants. The estimates in specifications II and IV pertain to all plants. Both the model with wage adjustment only (specifications I and II), and the model with wage and employment adjustment (specifications III and IV) are estimated.[27]

[27] In all specifications in Table 3, the top and bottom 1% of the plant level distributions of average wage, wage bill-to-revenue ratio, and labor productivity were trimmed to reduce the influence of some major outliers. The total employment of a plant was also restricted to the range 5 to 10,000 employees.

In all specifications in Table II, the estimates for the production function parameters α and γ have the expected signs, and fall in the interval $(0, 1)$. The average estimated value of $\alpha + \gamma$ when revenue is measured by the value of shipments is about 0.37. When value added is used as the revenue measure in the bottom panel, the estimates of α and γ are much higher, and still highly significant. The estimates of $\alpha + \gamma$ average 0.72 across specifications. The estimates of γ are comparable and sometimes exceed α in the top and bottom panels in Table II. The importance of labor quality relative to quantity in production accords well with Griliches and Regev's (1995) and Bahk and Gort's (1993) findings that labor quality plays an important role in explaining productivity differences across plants.

In most specifications in Table II, the estimates of the adjustment cost parameters have the expected signs, and all are highly significant.[28] The estimates reveal asymmetry in adjustment costs. Except for one case (specification II with alternative growth rate), downward adjustment cost parameters are higher than upward adjustment cost parameters, both for wage and employment adjustment. The equality of the upward and downward adjustment cost parameters are rejected at high levels of significance across most specifications. Furthermore, downward adjustment cost parameters are generally larger in absolute value when all plants are considered (specifications II and IV), compared to the case of continuing plants only (specifications I and III). This difference suggests that failing plants may face steeper downward adjustment costs. The adjustment cost parameter estimates are generally smaller under alternative growth rates, potentially because the alternative growth rates are robust to remaining outliers in wage and employment growth.

[28] In two cases, the estimated upward adjustment cost parameters are negative, with small absolute values. These cases, which also emerge in some prior work with quadratic adjustment costs (e.g. Hall (2004)), may result from a combination of remaining outliers in the data and measurement error. An implicit constraint on the model's adjustment cost parameters is non-negativity. As a robustness check, the model was re-estimated subject to these constraints for the specifications where an adjustment cost parameter estimate was negative in Table II. If the model's specification is not largely at odds with the data, moving from the unconstrained to the constrained estimation should not result in drastic sign and significance changes in other parameters. As expected, the constrained estimates were zero for the negative parameter estimates in Table II, and the other parameter values were very similar to those in Table II.

5.2 The magnitude of adjustment costs

Because the percent adjustments in wages and employment apply to different bases in various adjustment cost specifications, it is not appropriate to simply compare the estimated magnitudes of λ^U and ν^U, or λ^D and ν^D, to gauge the relative importance of different adjustment costs. The approach here is to assess the cost of adjustment for wages versus employment by comparing the shares of the average wage and employment adjustment costs in a plant's initial revenue over a 5-year window between two censuses. The adjustment costs are calculated over a 5-year period between two census years t and $t+5$, whereas the current revenue is calculated for census year t.

Table III presents the quartiles of the revenue shares of adjustment costs based on the parameter estimates in Table II. Rather than picking a certain specification arbitrarily, for each adjustment cost parameter its maximum estimate across all specifications for the case of total value of shipments in Table II is used to obtain the estimates in Table III. This approach provides an upper bound on the revenue share of adjustment costs. Panel (a) considers estimates from the models using continuing plants only, corresponding to specifications I and III in Table II. For the model with wage adjustment only, at the median adjustment cost, the adjustment cost is 0.8% of revenue when all directions of adjustment are considered together, and around 0.4% and 2% when upward and downward adjustments are considered, respectively. For the model with both wage and employment adjustment, wage adjustment cost is about 1.0% of revenue at the median wage adjustment cost, and upward wage adjustment cost constitutes about 0.6% of revenue, while downward wage adjustment cost makes up about 1.6% of revenue. Employment adjustment cost has a significant share (nearly 4%) of revenue at the median employment adjustment cost when downward employment adjustments are considered. For upward employment adjustments, this share is about 3% at the median employment adjustment cost.

A similar pattern emerges in panel (b) of Table III, which focuses on the estimated parameters using all plants, corresponding to specifications II and IV in Table II. For the model with wage adjustment only, upward and downward wage adjustment costs make up, respectively, about 0.8% and 3.5% of revenue at the median wage adjustment cost. For the model with both adjustment margins, these shares are about 0.1% and 3.2%, respectively. For employment ad-

justment, upward adjustment cost is about 5.9% of revenue, whereas the downward adjustment cost constitutes about 8.2% of revenue, both measured at the median employment adjustment cost.

Adjustment costs constitute a much larger fraction of revenue when only the exiting plants are considered, as panel (c) of Table III indicates. For almost all model specifications and for both upward and downward adjustment, the shares in panel (c) are much higher compared to panels (a) and (b). These magnitudes suggest that downward adjustment costs can potentially be a substantial burden for plants, especially for those nearing exit.

How much do adjustment costs in wages change over the life-cycle of a plant? The evolution of adjustment costs along the plant life-cycle is depicted in Figure III for continuing plants. These estimates are based on the largest adjustment cost parameter estimates from the model with wage adjustment cost only using all plants, corresponding to specifications labelled II in the left panel of Table II, which also generate the left side of panel (b) in Table III. For plants that are five years from their first census, wage adjustment cost makes up about 2.3% of revenue annually at the median. This share increases slightly as new plants age and then declines to just below 1.6% by the time a plant is 20 years away from entry. Somewhat reversal of this pattern emerges for plants approaching exit. Wage adjustment cost grows to about 2.2% of a plant's revenue five years before its exit, doubling from about 1.1% when a plant stands twenty years from exit.

5.3 Estimation using annual data

The estimates for annual adjustment costs were obtained using data in CM, which has quinquennial frequency. Longer time period between two consecutive observations of a plant could lead to lower adjustment cost estimates, as plants may have more flexibility to adjust over a longer horizon. If adjustment costs are highly convex, plants would have an incentive to spread adjustments over time. The estimation is repeated using annual frequency data from the US Census Bureau's Annual Survey of Manufactures (ASM) for the period 1972-2009. As in the case of the CM, plants were linked over time. The unbalanced panel constructed from the ASM has a number of important differences from the panel from the CM. First, ASM pertains

mainly to plants that are relatively large.[29] The sample is thus subject to non-random selection. Furthermore, a plant may enter to or drop from the ASM panel simply because its size changes to a level that is above or below the threshold of inclusion with certainty in the ASM. Second, the sampling frame for the ASM changes every five years, implying that some plants may enter, whereas others may be dropped from the sample over time. Therefore, actual plant entry/exit do not necessarily coincide with entry to/exit from the sample. The results based on the ASM thus may differ from the ones based on the CM for reasons in addition to the differences in the frequency of observation.[30]

The results from the GMM estimation based on the ASM panel are in Table A.II. Sample weights in the ASM were used to obtain the population estimates. The results appear to be qualitatively similar across Tables II and A.II. Most importantly, downward adjustment costs are still larger than upward adjustment costs, and upward adjustment costs are generally small for both wages and employment. However, there are a few important differences. First, in Table A.II estimates of production function parameters α and γ are generally smaller. This discrepancy could be driven by the fact that the plants in the sample used for the results in Table A.II are much larger, and may have different production technologies compared to smaller ones that are in the sample used for the results in Table II. Second, downward employment adjustment cost parameter estimates are much larger compared to those in Table II, whereas downward wage adjustment cost parameter estimates have comparable magnitudes. These patterns suggest that downward adjustment costs may be higher in the short term for employment. Overall, the estimates indicate that the frequency of adjustment may not be entirely driving the qualitative results obtained with the quinquennial data based on the CM.

How do the estimates compare with some of the prior estimates? There does not exist a sizeable prior literature on the estimates of wage adjustment costs. However, several studies

[29] The Annual Survey of Manufactures (ASM) is a sample survey of approximately 50,000 establishments. A new sample is selected at 5-year intervals beginning the second survey year subsequent to CM. Large plants are sampled with certainty, and smaller plants are sampled according to a stratified random sampling with probabilities that vary with plant size. The stratification is done across finely defined manufacturing industries. For details on the sampling procedure used in ASM, see http://www.census.gov/manufacturing/asm/how_the_data_are_collected/index.html.

[30] A minimum plant size requirement of 50 employees is imposed to ensure that the estimation is not unduly influenced by smaller plants.

have provided estimates of labor adjustment costs. These studies span a variety of model structures, samples, and aggregations, making the comparison inherently difficult. Nevertheless, most studies find positive labor adjustment costs, based on different specifications. Adjustment costs are positive and significant in studies using a quadratic or higher-order specification with or without interactions among different adjustment margins.[31] The estimates of labor adjustment costs in this paper are within the range of the estimates found in these previous studies, though there is strong evidence of asymmetric adjustment costs here. The estimates indicate relatively large downward employment adjustment costs compared to upward adjustment costs. Furthermore, as shown in Table III, wage adjustment costs are smaller when wage adjustment costs are estimated together with employment adjustment costs, suggesting that not controlling for labor quantity adjustments may introduce bias in adjustment cost estimates for labor quality, and vice versa.

6 Conclusion

This paper investigated the dynamics of wages along the life-cycle of manufacturing plants. Some stylized facts emerge. New plants start with a lower average wage compared to mature plants. A surviving plant's average wages increases and catches up with those of mature plants. The average wage of a plant approaching exit falls, but not as steeply as it rises in the case of surviving new plants. Wage bill constitutes a lower fraction of young plants' revenue because their labor productivity, as measured by revenue per worker, grows faster than its average wage. For plants approaching exit, average wage does not fall as fast as labor productivity does, implying that a higher fraction of revenue must be dedicated to wage bill.

A model of plant-level dynamics was introduced to explore the potential role of adjustment costs in wages. In the model, changes in average wage result from both changes in average labor quality and changes in the exogenous wage rate per unit of quality. The estimated parameters of the model reveal evidence of asymmetric adjustment costs for average wage. The estimated upper bounds on revenue share of wage adjustment costs indicate economic significance of

[31] E.g. Shapiro (1986), Pfann and Palm (1993), Alonso-Borrego (1998), and Merz and Yashiv (2007)), with the exception of Hall (2004), who finds quadratic costs that are not significantly different from zero using data at high-level industry aggregations.

such costs at the plant level, especially those associated with downward adjustment. The estimates remain economically significant when wage adjustment cost is considered jointly with employment adjustment cost.

The cost of adjusting average wage downward may increase the likelihood of exit and speed up the demise of failing plants. Further research can quantify the importance of these costs in exit. Such effects can potentially be larger in the case of European plants, which face stricter labor regulations than in the U.S.. The movements in average wage at the plant level can also be decomposed further. A challenging and promising avenue is to separate the cost of adjustments in response to changes in the wage rate per unit of labor quality from the cost of adjustments due to changes in average labor quality only. Finally, the coevolution of average wage and labor quality may also point to a theory of plant life-cycle where new plants start with a simple worker hierarchy that progressively becomes complex with the addition of workers of higher quality, and where plant decline triggers a sluggish dismantling of this hierarchy and slow reduction in labor costs because of potentially high costs of dispensing with certain type of workers.

References

[1] Abraham, Arpad, and White, Kirk (2006) "The Dynamics of Plant Level Productivity in U.S. Manufacturing", Center for Economic Studies Working Paper No. 06-20.

[2] Alonso-Borrego, Cesár (1998) "Demand for Labour Inputs and Adjustment Costs: Evidence from Spanish Manufacturing Firms", *Labour Economics*, **5**: 475-497.

[3] Bahk, Byong-Hyong and Gort, Michael (1993) "Decomposing Learning by Doing in New Plants", *Journal of Political Economy*, **101**: 561-583.

[4] Brown, Charles, and Medoff, James L. (1989) "The Employer Size-Wage Effect", *Journal of Political Economy*, **97**: 1027-1059.

[5] Brown, Charles, and Medoff, James L. (2003) "Firm Age and Wages", *Journal of Labor Economics*, **21**: 677-698.

[6] Foster, Lucia, John, Haltiwanger, and Syverson, Chad (2008) "Reallocation, Firm Turnover, and Efficiency: Selection on Productivity or Profitability?", *American Economic Review*, **98**: 394-425.

[7] Fuss, Catherine and Wintr, Ladislav (2009) "Rigid Labour Compensation and Flexible Employment? Firm-level Evidence with Regard to Productivity for Belgium", Research series 200903-11, National Bank of Belgium.

[8] Gort, Michael, Bahk, Byong-Hyong, and Wall, Richard (1993) "Decomposing Technical Change", *Southern Economic Journal*, **62** (July 1993).

[9] Gort, Michael, Sapra, Sunil, and Bahk, Byong-Hyong (1990) "Old Inputs, New Inputs, and Productivity", in 1990 Census Research Conference, Washington: Bureau of the Census.

[10] Griliches, Zvi, and Regev, Haim (1995) "Firm Productivity in Israeli Industry 1979-1988", *Journal of Econometrics*, **65**: 175-203.

[11] Hall, Robert E. (2004) "Measuring Factor Adjustment Costs," *Quarterly Journal of Economics*, **119**: 899-927.

[12] Hansen, Lars P. and Singleton, Kenneth J. (1982) "Generalized Instrumental Variables Estimation of Nonlinear Rational Expectations Models", *Econometrica*, **50**: 1269-1286.

[13] Heyman, Fredrik (2007) "Firm Size or Firm Age? The Effect on Wages Using Matched Employer-Employee Data", *Labour*, **21**: 237-263.

[14] Kölling, Arnd, Schnabel, Claus, and Wagner, Joachim (2002) "Establishment Age and Wages: Evidence from German Linked Employer-Employee Data", Friedrich-Alexander University Erlangen-Nuremberg, Discussion Paper No. 13.

[15] Merz, Monika, and Yashiv, Eran (2007): "Labor and the Market Value of the Firm," *American Economic Review*, **97**: 1419–1431.

[16] Pakes, Ariel (1994) "The Estimation of Dynamic Structural Models: Problems and Prospects, Part II. Mixed Continuous-Discrete Control Models and Market Interactions", Chapter 5, pp. 171-259, of Advances in Econometrics: Proceedings of the 6th World Congress of the Econometric Society, edited by J.J. Laffont and C. Sims.

[17] Shapiro, M. (1986): "The Dynamic Demand for Labor and Capital", *Quarterly Journal of Economics*, **101**: 513–542.

[18] Stokey, Nancy L. and Lucas, Robert E. (1989) *Recursive Methods in Economic Dynamics.* Harvard University Press.

Table I. OLS estimation results for the evolution of average wage, labor productivity, and wages to revenue ratio

Dependent Variable:	**Average Wage ($1,000)**		**Revenue/Employment ($1,000)**		**Wage Bill/Revenue (%)**	
Independent Variables	**I**	**II**	**I**	**II**	**I**	**II**
Exit	-2.031***	-1.846***	-9.852***	-8.962***	0.675***	0.598***
	[0.018]	[0.018]	[0.160]	[0.157]	[0.038]	[0.038]
5 years to exit	-1.010***	-0.897***	-5.220***	-4.939***	0.634***	0.635***
	[0.019]	[0.019]	[0.174]	[0.171]	[0.042]	[0.041]
10 years to exit	-0.695***	-0.613***	-3.920***	-3.914***	0.461***	0.497***
	[0.024]	[0.024]	[0.219]	[0.215]	[0.053]	[0.052]
15 years to exit	-0.447***	-0.412***	-2.566***	-3.010***	0.273***	0.369***
	[0.031]	[0.030]	[0.275]	[0.269]	[0.066]	[0.065]
20 years to exit	-0.159***	-0.179***	-1.240***	-2.281***	0.031	0.211**
	[0.038]	[0.038]	[0.347]	[0.339]	[0.083]	[0.082]
Entry	-2.714***	-2.356***	-5.668***	-1.998***	-1.733***	-2.236***
	[0.017]	[0.017]	[0.156]	[0.155]	[0.037]	[0.038]
5 years from entry	-1.782***	-1.514***	-2.228***	0.282*	-1.247***	-1.580***
	[0.019]	[0.019]	[0.170]	[0.168]	[0.041]	[0.041]
10 years from entry	-1.180***	-0.964***	-1.141***	0.571**	-0.653***	-0.865***
	[0.023]	[0.023]	[0.207]	[0.204]	[0.050]	[0.049]
15 years from entry	-0.628***	-0.469***	-0.151	0.759***	-0.219***	-0.312***
	[0.026]	[0.026]	[0.239]	[0.234]	[0.057]	[0.057]
20 years from entry	-0.229***	-0.120**	1.537***	1.634***	-0.181**	-0.152
	[0.044]	[0.044]	[0.397]	[0.389]	[0.095]	[0.094]
Industry x year fixed effects	Y	Y	Y	Y	Y	Y
Other controls	N	Y	N	Y	N	Y
N	1,219,769	1,219,769	1,219,769	1,219,769	1,219,769	1,219,769
R^2	0.29	0.30	0.38	0.41	0.43	0.44

Notes: Standard errors in brackets. (*),(**),(***) indicate significance at 10%,5%,1%, respectively. Other controls include a cubic spline in plant size (measured by total employment) and an indicator of multi-plant firm. The omitted category is the plants that are more than twenty years away from their entry or exit, and the plants whose entry or exit points do not fall into the sample period.

Table II. GMM estimates for the model's parameters using CM sample

	Adjustment costs (conventional growth rate)				Adjustment cost (alternative growth rate)			
	wage adjustment		employment adjustment		wage adjustment		employment adjustment	
Parameter	Continuing plants (I)	All plants (II)	Continuing plants (III)	All plants (IV)	Continuing plants (I)	All plants (II)	Continuing plants (III)	All plants (IV)
Revenue measure: Total value of shipments (deflated)								
α	0.12*** [0.0005]	0.11*** [0.0006]	0.12*** [0.002]	0.09*** [0.0006]	0.12*** [0.0004]	0.13*** [0.0004]	0.12*** [0.0005]	0.14*** [0.001]
γ	0.19*** [0.0004]	0.18*** [0.0005]	0.25*** [0.002]	0.24*** [0.0007]	0.27*** [0.0009]	0.42*** [0.002]	0.18*** [0.002]	0.13*** [0.002]
λ^U	2.89*** [0.05]	2.62*** [0.04]	10.73*** [0.41]	-0.98*** [0.01]	5.94*** [0.11]	14.18*** [0.15]	1.25*** [0.03]	2.77*** [0.08]
λ^D	35.61*** [0.25]	63.10*** [0.42]	20.81*** [0.74]	56.48*** [0.33]	9.55*** [0.07]	6.30*** [0.06]	10.10*** [0.12]	10.46*** [0.33]
v^U	-	-	8.62*** [0.52]	0.56*** [0.03]	-	-	1.97*** [0.42]	13.43*** [0.72]
v^D	-	-	23.17*** [4.82]	42.91*** [1.03]	-	-	8.90*** [0.12]	88.21*** [1.94]
Revenue measure: Value added (deflated)								
α	0.36*** [0.0005]	0.39*** [0.0008]	0.25*** [0.003]	0.33*** [0.0007]	0.32*** [0.0002]	0.37*** [0.001]	0.27** [0.001]	0.39*** [0.003]
γ	0.47*** [0.002]	0.44*** [0.001]	0.37*** [0.004]	0.39*** [0.002]	0.35*** [0.0002]	0.27*** [0.001]	0.22*** [0.002]	0.42*** [0.002]
λ^U	1.39*** [0.04]	1.00*** [0.02]	3.73*** [0.24]	1.38*** [0.04]	3.44*** [0.08]	3.25*** [0.06]	1.04*** [0.03]	-0.54*** [0.05]
λ^D	15.63*** [0.17]	13.95*** [0.20]	29.32*** [0.47]	19.25*** [0.21]	10.31*** [0.06]	8.34*** [0.06]	6.82*** [0.08]	1.37*** [0.22]
v^U	-	-	3.46*** [0.28]	1.13*** [0.04]	-	-	2.03*** [0.32]	17.74*** [0.79]
v^D	-	-	41.42*** [2.70]	47.60*** [0.72]	-	-	6.69*** [0.09]	58.32*** [2.20]
N	804,245	986,977	804,245	986,977	804,245	986,977	804,245	986,977

Notes: Standard errors in brackets. (*),(**),(***) indicate significance at 10%,5%,1%, respectively.

Table III. Adjustment costs as a percentage of revenue

(a) Estimates using continuing plants only

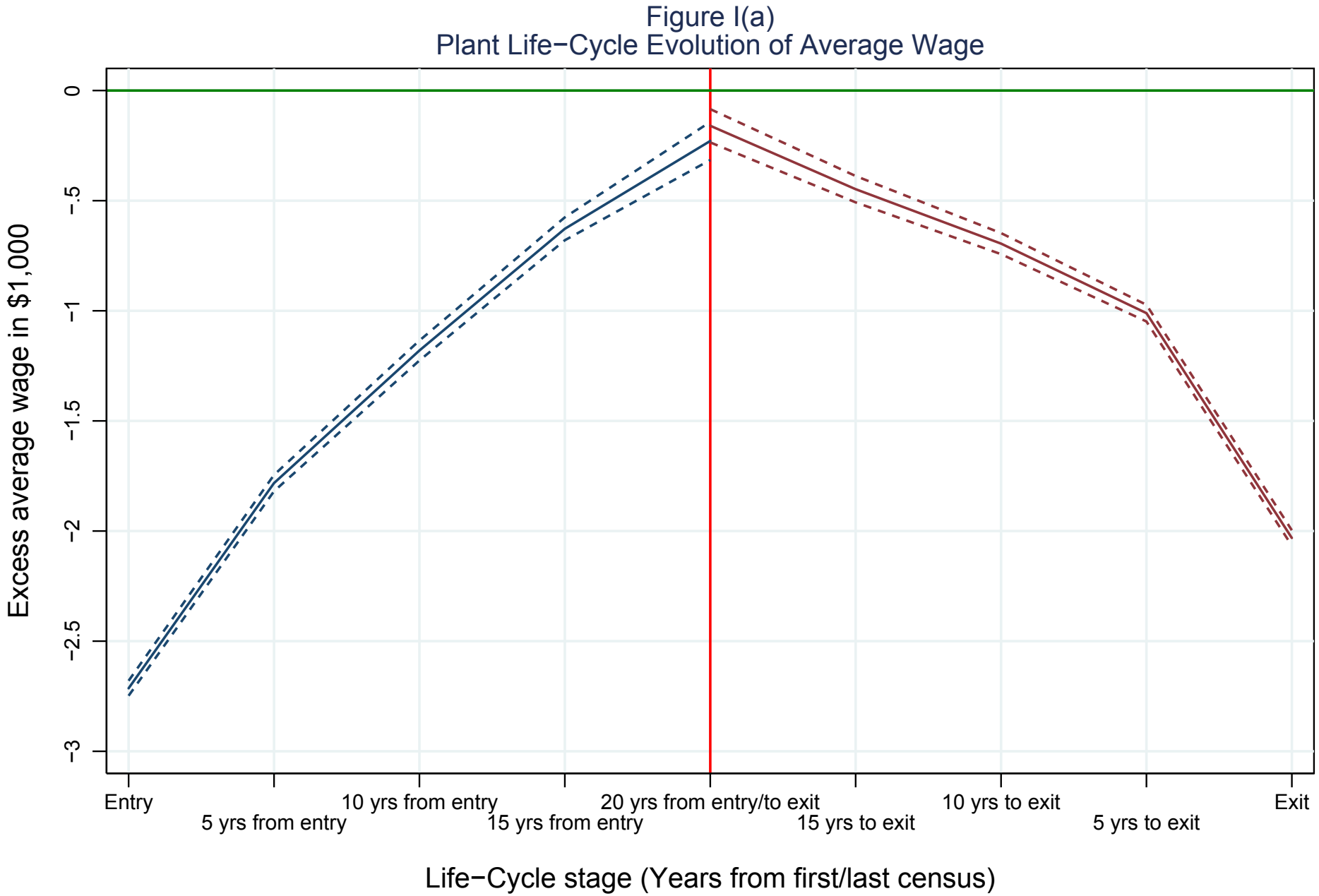
		Model with wage adjustment only			Model with wage and employment adjustment				
	Adjustment Cost Percentile	All Upward		Downward	All	Upward (Wages)	Downward (Wages)	Upward (Employment)	Downward (Employment)
Wage Adjustment	25	0.1%	0.1%	0.4%	0.2%	0.1%	0.3%		
	50	0.8%	0.4%	2.0%	1.0%	0.6%	1.6%		
	75	3.7%	1.5%	7.6%	4.2%	2.7%	6.2%		
Employment Adjustment	25				0.6%			0.6%	0.5%
	50				3.2%			2.9%	3.8%
	75				14.3%			12.6%	16.9%

(b) Estimates using all plants: Continuing plants only

		Model with wage adjustment only			Model with wage and employment adjustment				
	Adjustment Cost Percentile	All Upward		Downward	All	Upward (Wages)	Downward (Wages)	Upward (Employment)	Downward (Employment)
Wage Adjustment	25	0.3%	0.2%	0.7%	0.0%	0.0%	0.6%		
	50	1.6%	0.8%	3.5%	0.4%	0.1%	3.2%		
	75	7.4%	3.5%	13.4%	3.4%	3.9%	2.9%		
Employment Adjustment	25				1.2%			1.2%	1.1%
	50				6.7%			5.9%	8.2%
	75				29.9%			26.0%	35.9%

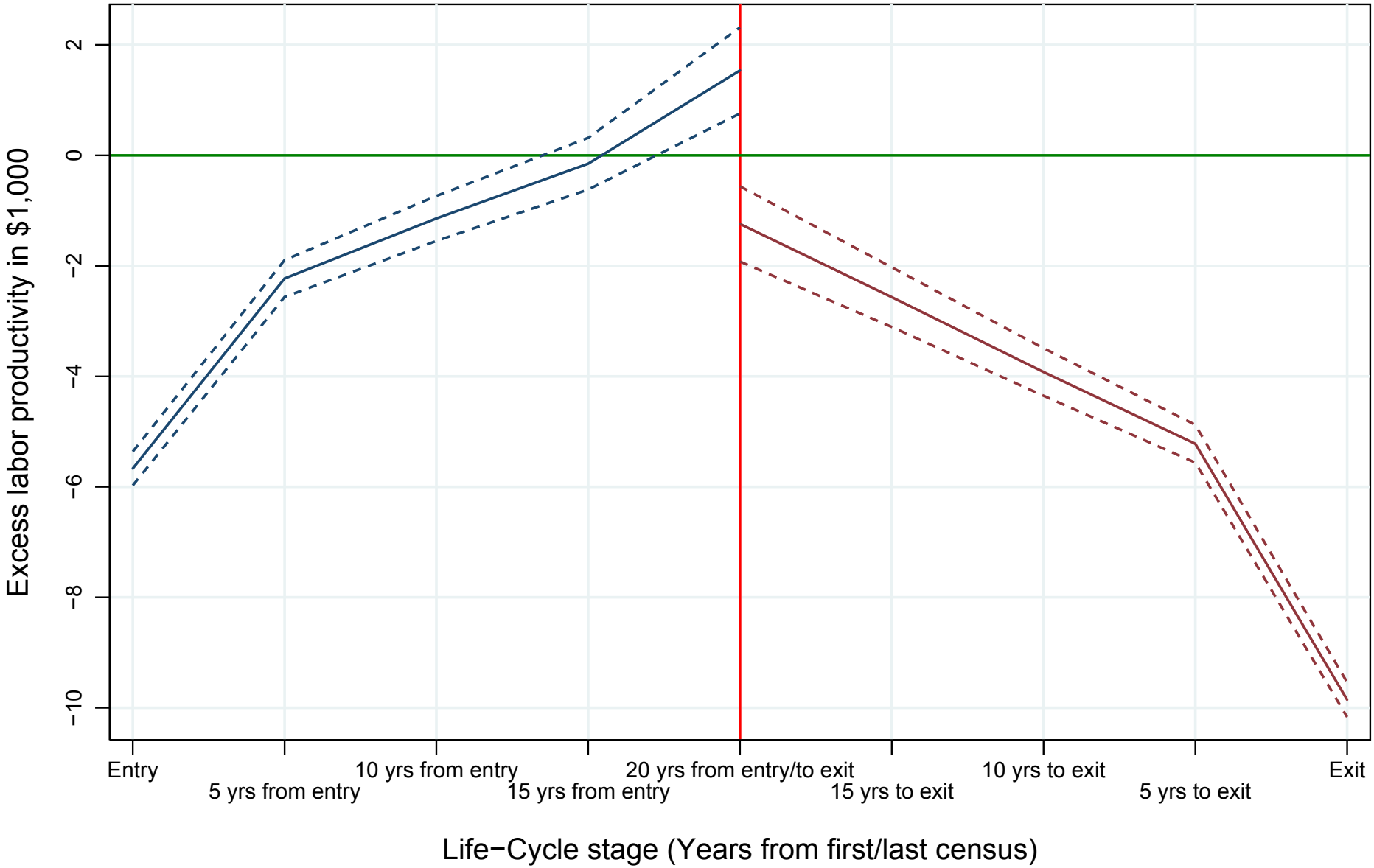
(c) Estimates using all plants: Exiting plants only

		Model with wage adjustment only			Model with wage and employment adjustment				
	Adjustment Cost Percentile	All Upward		Downward	All	Upward (Wages)	Downward (Wages)	Upward (Employment)	Downward (Employment)
Wage Adjustment	25	0.5%	0.2%	1.3%	0.1%	0.0%	1.1%		
	50	3.0%	1.2%	6.7%	0.9%	0.1%	6.0%		
	75	14.2%	5.3%	25.7%	7.9%	0.5%	23.0%		
Employment Adjustment	25				2.4%			1.8%	3.4%
	50				14.8%			10.0%	22.4%
	75				70.6%			48.5%	94.7%



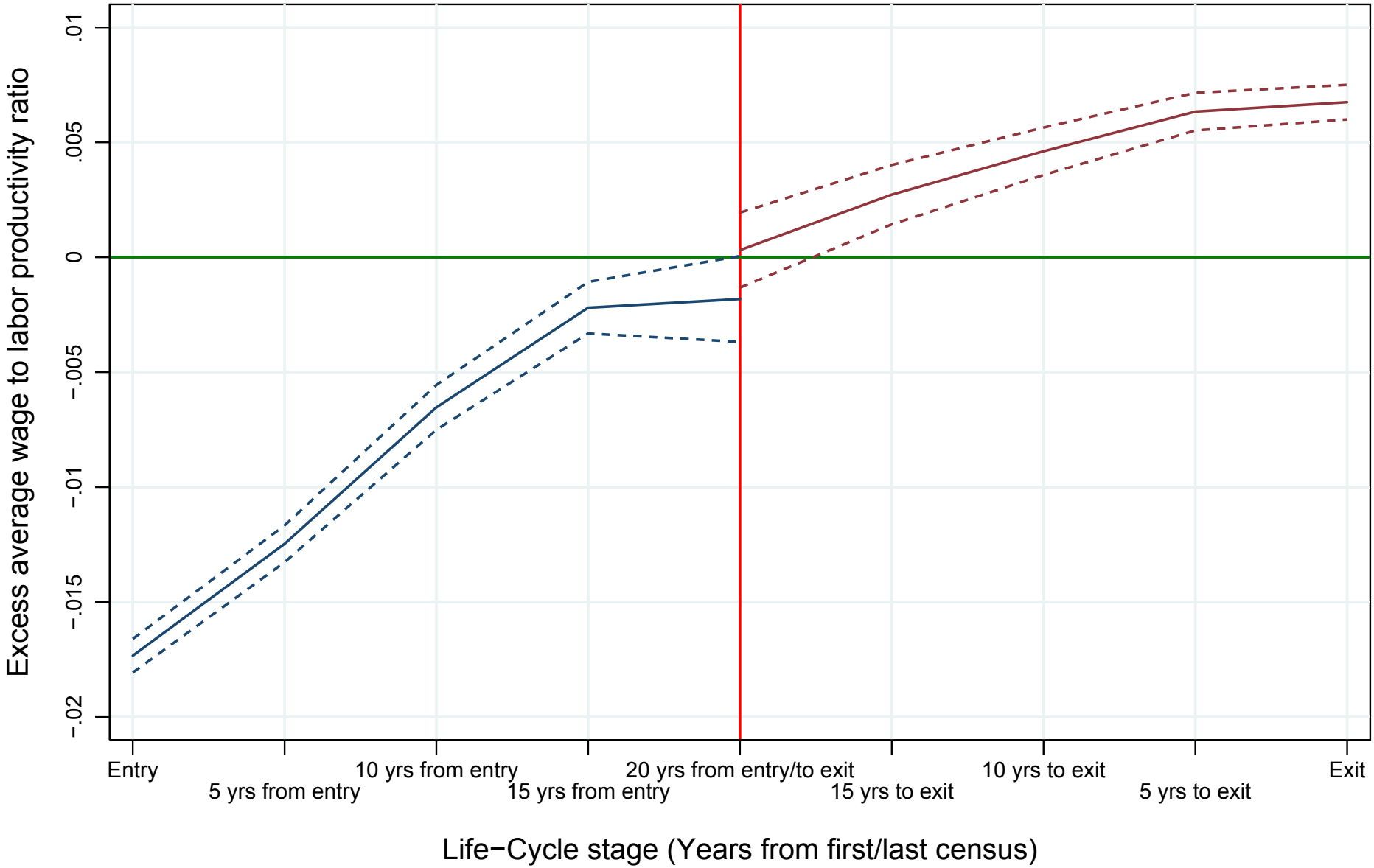
Notes: The horizontal line at zero represents the normalized value for the omitted category. Dashed lines are +/− 2 std. errors.

Figure I(b)
Plant Life−Cycle Evolution of Labor Productivity



Notes: The horizontal line at zero represents the normalized value for the omitted category. Dashed lines are +/− 2 std. errors.

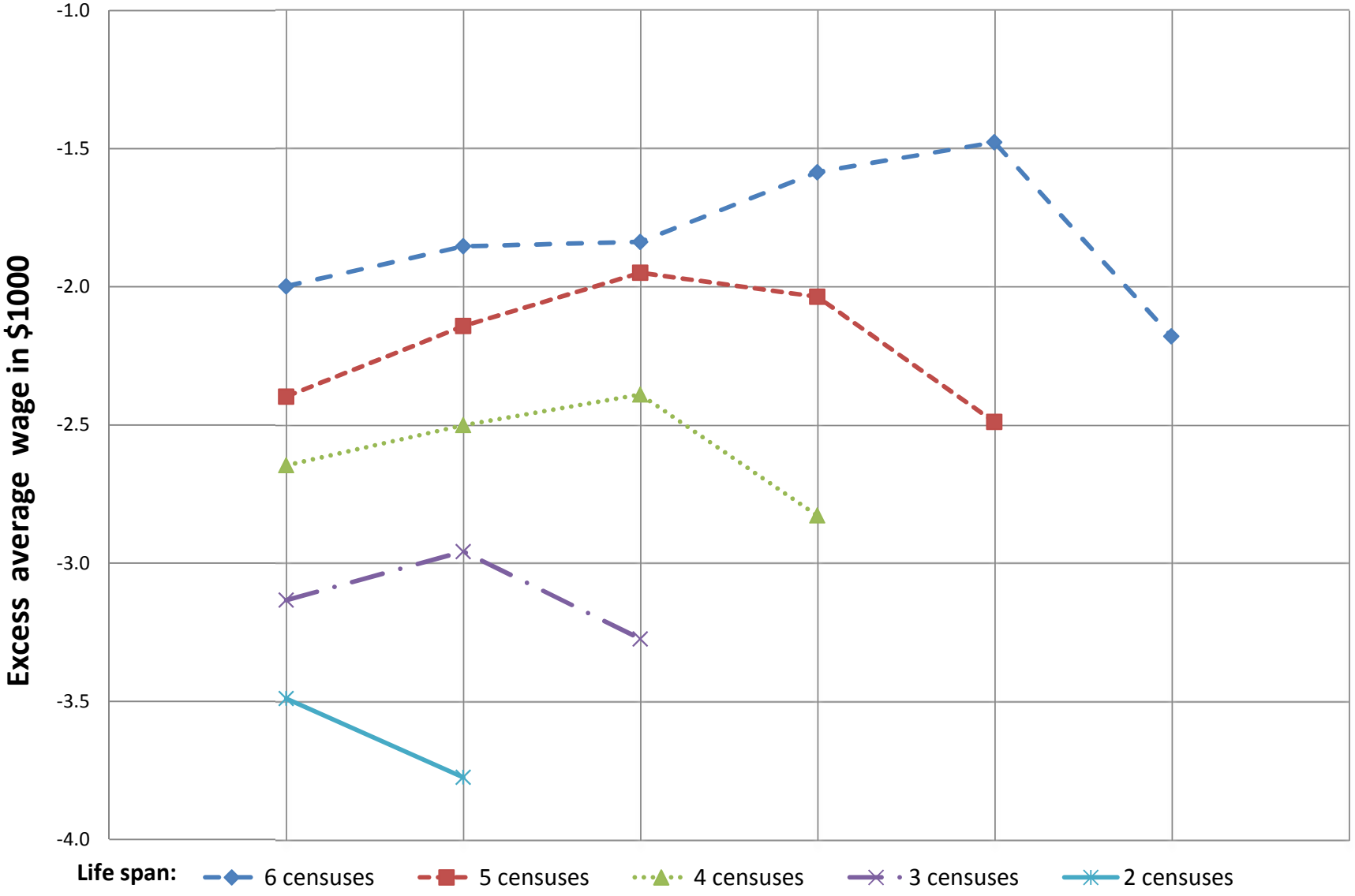
Figure I(c)
Plant Life−Cycle Evolution of Average Wage to Labor Productivity Ratio



Notes: The horizontal line at zero represents the normalized value for the omitted category. Dashed lines are +/− 2 std. errors.

Figure II

Plant life-cycle evolution of average wage by life span

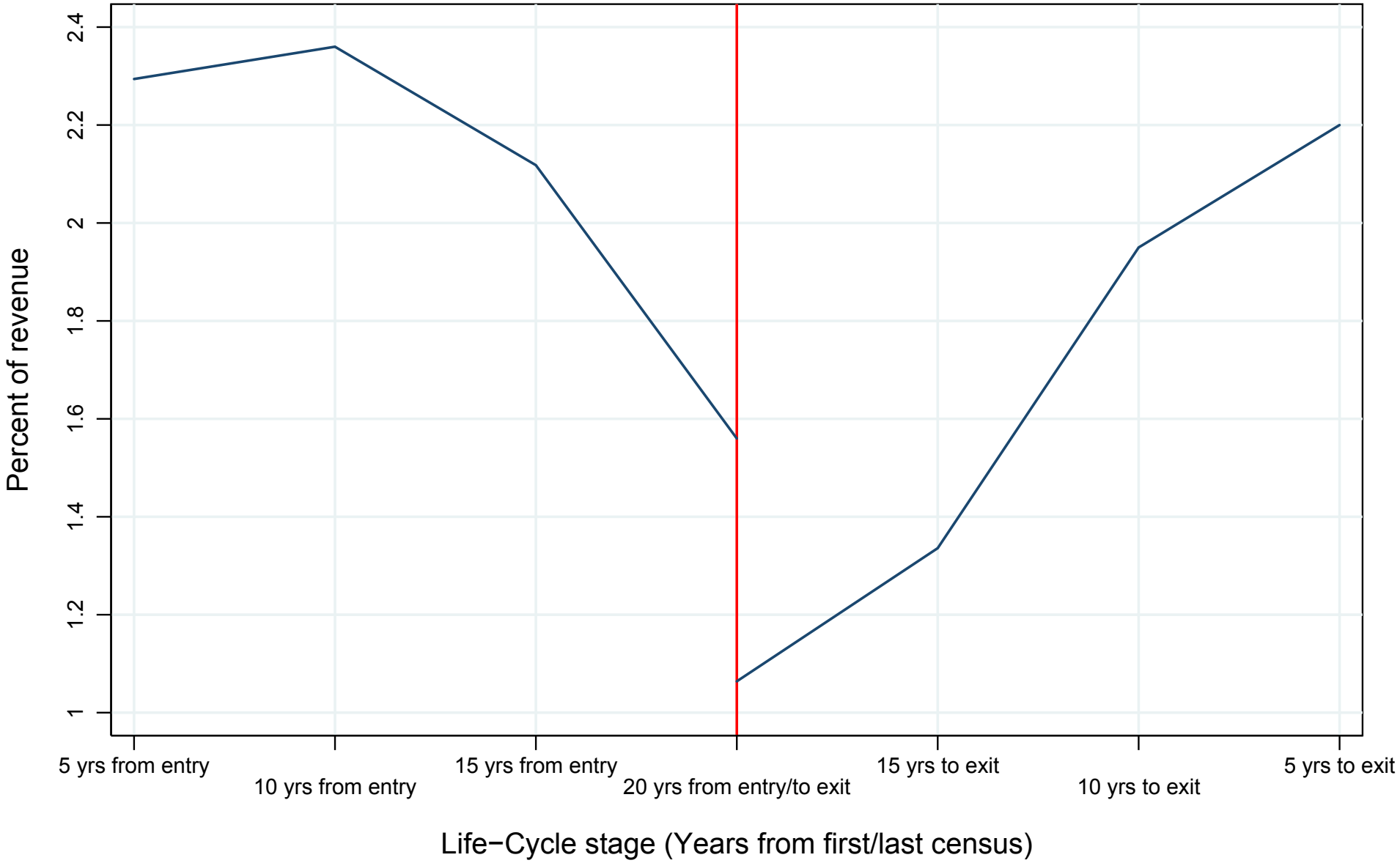


Notes: Excess average wage is measured with respect to the omitted category.

Figure III

Plant Life−Cycle Evolution of Wage Adjustment Cost as a Share of Revenue

Estimates based on the model with wage adjustment −− continuing plants only



Appendix

Table A.I. OLS estimation results for the evolution of average wage, labor productivity, and wages to revenue ratio by plant-life span

Dependent Variable:	Average Wage ($1,000)		Revenue/Employment ($1,000)		Wage Bill/Revenue (%)	
Independent Variables	**I**	**II**	**I**	**II**	**I**	**II**
Lifespan = 2						
Entry	-3.490***	-3.042***	-9.638***	-5.740***	-1.159***	-1.661***
	[0.035]	[0.035]	[0.315]	[0.310]	[0.076]	[0.075]
Exit	-3.775***	-3.356***	-12.241***	-8.871***	-0.692 ***	-1.111***
	[0.035]	[0.035]	[0.315]	[0.310]	[0.076]	[0.075]
Lifespan = 3						
Entry	-3.134***	-2.713***	-8.227***	-4.762***	-1.125***	-1.561**
	[0.047]	[0.047]	[0.425]	[0.417]	[0.102]	[0.101]
5 years from entry	-2.958***	-2.570***	-8.118***	-5.077***	-0.651***	-1.025***
	[0.047]	[0.047]	[0.423]	[0.415]	[0.101]	[0.101]
Exit	-3.274***	-2.920***	-11.227***	-9.112***	-0.131	-0.362***
	[0.048]	[0.048]	[0.436]	[0.427]	[0.104]	[0.104]
Lifespan = 4						
Entry	-2.646***	-2.283***	-6.716***	-4.280***	-0.883***	-1.158***
	[0.065]	[0.064]	[0.584]	[0.572]	[0.208]	[0.139]
5 years from entry	-2.501***	-2.157***	-5.009***	-3.662***	-0.773***	-1.043 ***
	[0.064]	[0.064]	[0.579]	[0.567]	[0.139]	[0.138]
10 years from entry	-2.390***	-2.078***	-7.191***	-5.444***	-0.132***	-0.305***
	[0.064]	[0.064]	[0.578]	[0.566]	[0.138]	[0.137]
Exit	-2.828***	-2.547**	-11.091***	-10.172***	0.391**	0.354**
	[0.066]	[0.065]	[0.592]	[0.579]	[0.142]	[0.140]
Lifespan = 5						
Entry	-2.397***	-2.081***	-6.466***	-4.957***	-0.838**	-0.967**
	[0.096]	[0.096]	[0.868]	[0.849]	[0.208]	[0.206]
5 years from entry	-2.142***	-1.824***	-5.009***	-3.140***	-1.003***	-1.196***
	[0.096]	[0.095]	[0.863]	[0.845]	[0.207]	[0.205]
10 years from entry	-1.949***	-1.665***	-5.775***	-4.353***	-0.064	-0.192
	[0.095]	[0.094]	[0.854]	[0.835]	[0.205]	[0.203]
15 years from entry	-2.036***	-1.469***	-7.734***	-7.087***	0.328**	0.323*
	[0.077]	[0.077]	[0.698]	[0.683]	[0.167]	[0.166]
Exit	-2.489***	-2.120**	-10.442***	-10.741***	0.246	-0.404
	[0.141]	[0.140]	[1.267]	[1.240]	[0.304]	[0.301]

Table A.I. Continued

Lifespan = 6

Entry	-1.998***	-1.763***	-2.932**	-2.692***	-0.458	-0.964***
	[0.166]	[0.165]	[1.497]	[1.464]	[0.359]	[0.356]
5 years from entry	-1.853***	-1.581**	-2.053	-0.991	-0.896**	-0.151
	[0.166]	[0.165]	[1.501]	[1.468]	[0.360]	[0.094]
10 years from entry	-1.837***	-1.573***	-4.236**	-3.245**	-0.018	-0.077
	[0.164]	[0.163]	[1.480]	[1.448]	[0.355]	[0.351]
15 years from entry	-1.585***	-1.367**	-2.240*	-2.192	-0.062	-0.004
	[0.162]	[0.161]	[1.458]	[1.426]	[0.349]	[0.346]
20 years from entry	-1.477***	-1.287**	-3.559**	-3.889***	0.203	0.351
	[0.166]	[0.165]	[1.494]	[1.461]	[0.358]	[0.354]
Exit	-2.179***	-1.998***	-6.112***	-6.792***	0.181	0.374
	[0.167]	[0.165]	[1.502]	[1.469]	[0.360]	[0.356]
Industry x year fixed effects	Y	Y	Y	Y	Y	Y
Other controls	N	Y	N	Y	N	Y
N	1,219,769	1,219,769	1,219,769	1,219,769	1,219,769	1,219,769
R^2	0.29	0.30	0.38	0.41	0.43	0.44

Notes: Standard errors in brackets. (*),(**),(***) indicate significance at 10%,5%,1%, respectively. Life-span indicators were interacted with life-cycle indicators. Other controls include a cubic spline in plant size (measured by total employment) and an indicator of multi-plant firm. The omitted category is plants that are at

Table A.II. GMM estimates for the model's parameters using ASM sample

	Revenue measure: Total value of shipments (deflated)							
	Adjustment costs (conventional growth rate)				Adjustment cost (alternative growth rate)			
	wage adjustment		employment adjustment		wage adjustment		employment adjustment	
Parameter	Continuing plants (I)	All plants (II)	Continuing plants (III)	All plants (IV)	Continuing plants (I)	All plants (II)	Continuing plants (III)	All plants (IV)
α	0.11*** [0.00006]	0.11*** [0.00006]	0.10*** [0.002]	0.07*** [0.0001]	0.14*** [0.00007]	0.14*** [0.00006]	0.09*** [0.0004]	0.09*** [0.0001]
γ	0.17*** [0.00007]	0.17*** [0.00012]	0.22*** [0.00008]	0.20*** [0.0001]	0.21*** [0.0003]	0.31*** [0.0003]	0.18*** [0.0003]	0.12*** [0.0003]
λ^U	3.72*** [0.006]	1.72*** [0.004]	3.44*** [0.004]	0.83*** [0.004]	1.45*** [0.007]	1.32*** [0.008]	1.56*** [0.004]	1.31*** [0.008]
λ^D	1.31*** [0.008]	2.33*** [0.002]	2.76*** [0.009]	1.46*** [0.001]	18.82*** [0.052]	2.89*** [0.007]	19.73*** [0.049]	2.89*** [0.007]
v^U	-	-	0.64*** [0.21]	2.77*** [0.14]	-	-	1.53*** [0.538]	1.77*** [0.221]
v^D	-	-	96.39* [40.5]	32.65*** [6.0]	-	-	44.39*** [15.6]	43.49*** [8.5]
	Revenue measure: Value added (deflated)							
α	0.26*** [0.0002]	0.26*** [0.0002]	0.10*** [0.0005]	0.17*** [0.0002]	0.31*** [0.0001]	0.32*** [0.0001]	0.32*** [0.0001]	0.33*** [0.0003]
γ	0.20*** [0.0001]	0.19*** [0.0001]	0.19*** [0.0001]	0.19*** [0.0001]	0.16*** [0.0003]	0.17*** [0.0003]	0.13*** [0.0004]	0.17*** [0.0003]
λ^U	0.77*** [0.005]	1.14*** [0.003]	2.33*** [0.005]	0.33*** [0.001]	1.73*** [0.005]	1.17*** [0.006]	1.44*** [0.006]	1.21*** [0.004]
λ^D	1.93*** [0.008]	1.24*** [0.001]	2.88*** [0.008]	0.24*** [0.001]	12.16*** [0.04]	1.81*** [0.006]	14.39*** [0.041]	1.82*** [0.006]
v^U	-	-	0.62*** [0.221]	0.33*** [0.113]	-	-	1.52** [0.741]	1.48*** [0.255]
v^D	-	-	98.48*** [22.0]	32.84*** [6.2]	-	-	86.32*** [14.2]	37.10*** [7.2]
N	3,926,908	3,927,407	3,926,908	3,927,407	3,926,908	3,927,407	3,926,908	3,927,407

Notes: Standard errors in brackets. (*),(**),(***) indicate significance at 10%,5%,1%, respectively.
Number of observations reflect sampling weights used.